

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Mechanical Engineering) SEMESTER - 8 Winter 2025 (Supply.)

Course :Bachelor of Technology (Mechanical Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 8

Subject Code & Name: BTMEC802F - NON-CONVENTIONAL ENERGY RESOURCES

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

- 1 The beam part of solar radiation can be measured using: a. Speedometer, b. Pyrheliometer, c. Pyranometer, d. All the options
- 2 The material is required to control the nuclear chain reaction of nuclear power plant: a. Coolant, b. Control rods, c. Moderator, d. None
- 3 The concentrating collectors are used for: a. Lower temperature applications ($<100^{\circ}\text{C}$), b. Higher temperature applications, ($>100^{\circ}\text{C}$),, c. Both types of applications, d. None
- 4 The material over which radiation diffusely reflect in all the directions is called as: a. Transparent surface, b. Opaque surface, c. Rough surface, d. None of the above
- 5 Penetration of UV solar radiation can be effectively obstructed by which medium, a. Air, b. Water, c. Ozone, d. Vacuum
- 6 The SI unit of Solar radiation intensity or solar constant, I is: a. W/m^2 , b. $\text{W}/\text{m}^2\text{-K}^2$ c. hp, d. Pascal
- 7 Hydroelectric power plant require following form of energy to work: a. Potential Energy in terms of water head, b. Thermal Energy, c. Coal, d. Crude Oil
- 8 Ocean Thermal Energy Power Plant, is working on the principle of: a. Density Difference, b. Temperature Difference, c. Elevation Difference, d. None
- 9 Major source of renewable energy on Earth is: a. Wind Energy, b. Solar Energy, c. Biogas, d. Ocean Thermal
- 10 Darrius type wind turbine has rotating axis, a. Vertical, b. Horizontal, c. Inclined, d. No Axis
- 11 The parameters influence the performance of Biogas are, a. Temperature, b. Pressure, c. Ph Value, d. All options
- 12 Following parameters are required while designing a solar collector: a. Solar declination angle, b. Zenith Angle, c. Solar Constant, d. All the options

Q2. Solve the following.

- A) Explain the main components of a concentrating Solar collector with a neat Diagram. Also mention at least two basic points of differences between the Concentrating Collectors and Flat plate Collectors. 6
- B) Discuss in brief the working of following one of the power plants based on conventional sources of energy, as mentioned below: 1) Coal fueled power plant 2) Gas turbine power plant With the help of neat line schematic. 6
- Q3. Solve the following.
- A) Explain the instruments used for the measurement of Solar radiation, with the help of neat schematic and its working principle. 6
- B) Explain the working of Ocean Thermal Energy conversion system with the help of neat schematic of components and cycle of operation. 6
- Q4. Solve Any Two of the following.
- A) Discuss construction and working of concentrating collector systems, with the help of neat schematic. 6
- B) What is Biomass? Discuss the various forms of biomass. 6
- C) What are the environmental impacts of geothermal energy? What are the advantages of geothermal energy? 6
- Q5. Solve Any Two of the following.
- A) Describe about the single crystal Silicon Cell along with the principle of working of a Solar Cell, with neat schematics. Also list out the applications of PV system. 6
- B) What is a distillation process? Describe the construction and working of a Solar Still with the help of neat sketches 6
- C) Discuss the devices used for harnessing the tidal energy, with the help of neat sketches. 6
- Q6. Solve Any Two of the following.
- A) Explain about the solar radiation received at the Earth's surface, mentioning spectral distribution of extra-terrestrial solar radiation and schematic representation of the mechanisms of absorption and scattering and beam and diffuse radiation received at the Earth's surface. 6
- B) Write short notes on (i) Fuel Processor, and (ii) Microbial fuel cell 6
- C) Describe the energy storage technology in the form of (i) Flywheel and (ii) Super-capacitor, with the help of neat line schematics. 6

*** End ***